Learning Objectives

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1. Explain and summarize the mechanisms underlying short-term blood pressure regulation by baroreceptors, chemoreceptors, atrial reflexes, and the central nervous system-ischemic response.
2. Explain the mechanism underlying long-term blood pressure regulation via baroreceptors in juxtaglomerular apparatus, including step-by-step activation of renin-angiotensin-aldosterone system.
3. Interpret the integrated responses compensating for a drop in arterial blood pressure as a result of hypovolemic shock caused by acute bleeding.
4. Discuss clinical applications of blood pressure regulation concepts

Review of Definitions

Stroke Volume (SV): Volume of blood pumped by the L/R ventricle in a single heartbeat. Affected by contractility, afterload, and preload.

Cardiac Output (CO): Volume of blood the heart pumps through the circulatory system **per minute**. CO ~5 L/min.

CO = SV x HR

Total Peripheral Resistance (TPR): Amount of **resistance to blood flow** in the systemic circulation. Also referred to as systemic vascular resistance (**SVR**).

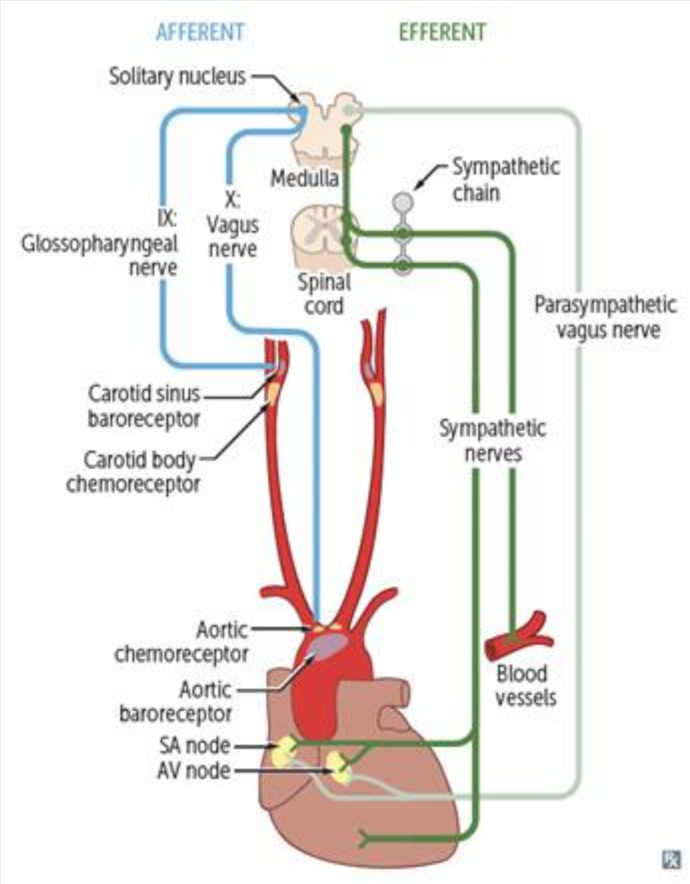
TPR = MAP / CO

Mean Arterial Pressure (MAP): The **average arterial pressure** throughout **one cardiac cycle**. MAP $\geq$ 60 mmHg to maintain adequate tissue perfusion.

MAP = CO x TPR

MAP = 2x (diastolic) + systolic / 3

Short-Term BP Regulation



DETECTORS	**PRIMARY SENSORS: BARORECEPTORS** • Detect stretch of vascular wall • Located in aortic arch & carotid sinus **SECONDARY SENSORS: CHEMORECEPTORS** • Detect changes in blood PO2, PCO2, and pH • Located in aortic arch & carotid sinus
AFFERENT	**CN IX (GLOSSOPHARYNGEAL NERVE)** • Pathway for impulses • Located in carotid sinus
AFFERENT	**CN X (VAGUS NERVE)** • Pathway for impulses • Located in aortic arch
CONTROL CENTER	**NUCLEUS TRACTUS SOLITARIUS (NTS)** • Receives impulses and responds • Located in medulla
EFFERENT	Pathway for CNS response to BP changes
EFFECTORS	• Pacemaker and muscle cells in the heart • Vascular smooth muscle cells in arteries and veins • Adrenal medulla

BARORECEPTOR Reflex - Mechanism

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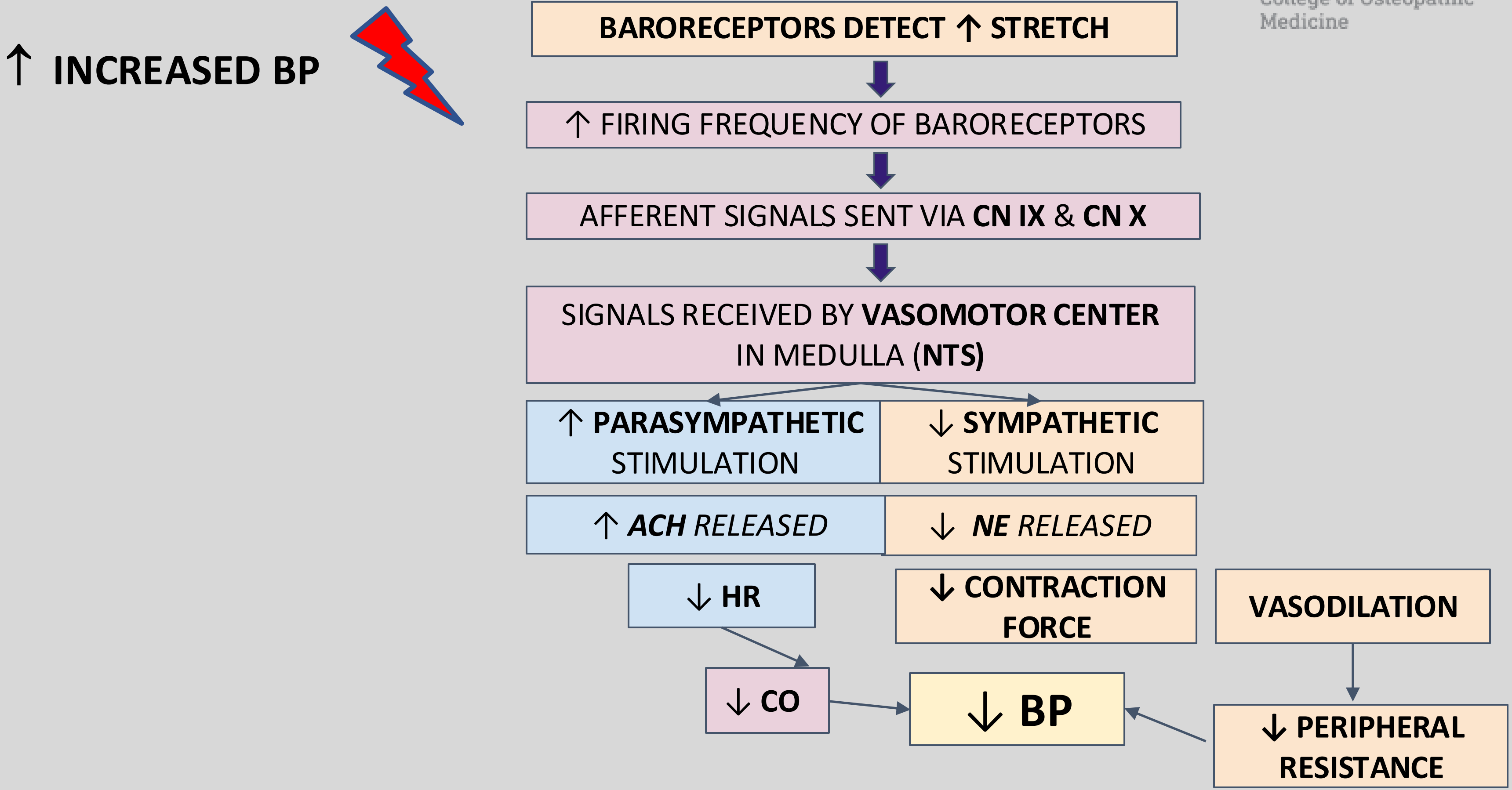

Baroreceptor Reflex - Mechanism

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↓ **DECREASED BP**

BARORECEPTORS DETECT ↓ STRETCH

↓ FIRING FREQUENCY OF BARORECEPTORS

AFFERENT SIGNALS SENT VIA **CN IX** & **CN X**

SIGNALS RECEIVED BY **VASOMOTOR CENTER** IN MEDULLA (**NTS**)

↓ **PARASYMPATHETIC** STIMULATION	↑ **SYMPATHETIC** STIMULATION
↓ ***ACH*** *RELEASED*	↑ ***NE*** *RELEASED*

↑ HR → **↑ CO**

↑ CONTRACTION FORCE

VASOCONSTRICTION → **↑ PERIPHERAL RESISTANCE**

↑ CO → **↑ BP** ← **↑ PERIPHERAL RESISTANCE**

Boron and Boulpaep, Medical Physiology, Updated 3rd ed., Elsevier, 2017 (Chapter 23)
Costanzo, Physiology 5th ed., Saunders/Elsevier, 2014 (Chapter 4: pp. 160-161)

SUMMARY - SHORT-TERM BP REGULATION - BARORECEPTORS

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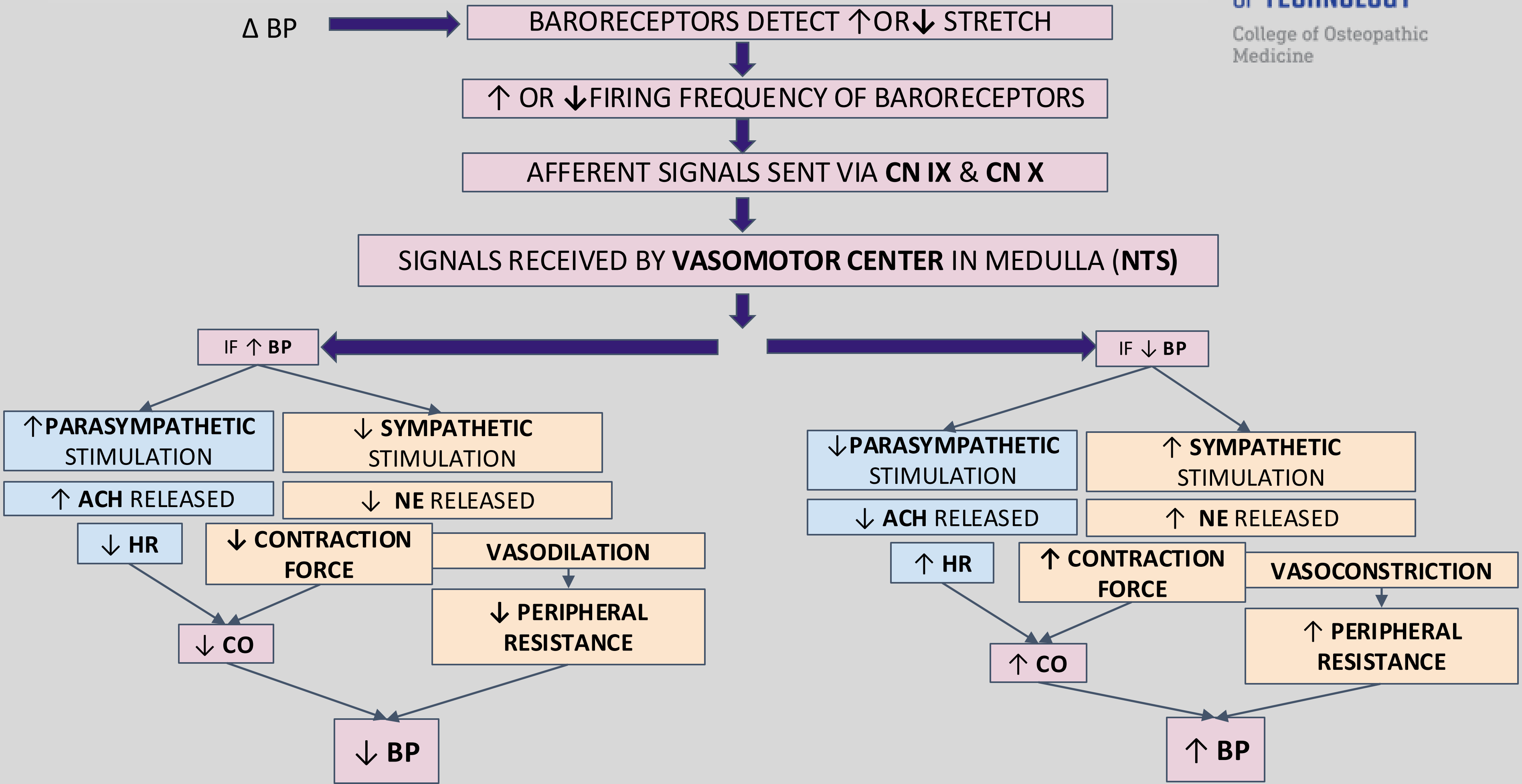


Chemoreceptors

- Specialized receptors, secondary sensors.
- Detect changes in blood O2, CO2, and pH.
- Types: **Peripheral** and **Central**

Mechanism

Peripheral chemoreceptors (in carotid & aortic bodies) detect **↑CO2, ↓ O2, ↓ pH**

⬇

Afferent signals sent via **CN IX** & **CN X**

⬇

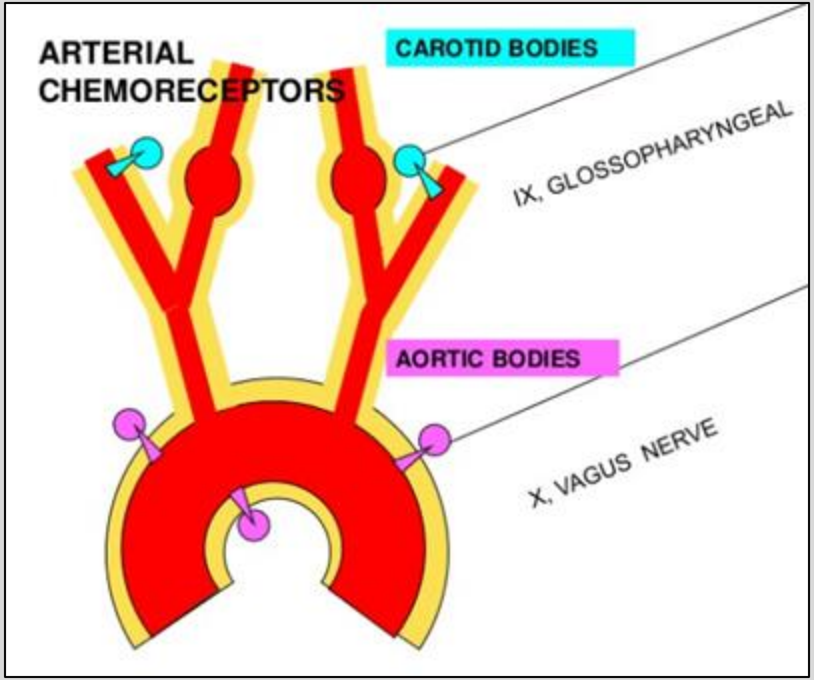
Stimulation of medullary respiratory center & **medullary VMC**

⬇

↑Sympathetic activity

⬇

↑BP



Atrial Reflex / BAINBRIDGE Reflex

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Physiologic reflex, mediated by **stretch receptors in the atria** (B-fibers)

Role of Reflex:

- Maintain normal blood volume by *attempting* to eliminate excessive fluid in the circulation = **maintain cardiovascular system homeostasis**.

Increased venous return to the heart results in atrial distension, leading to ↑HR

MECHANISM

↑BLOOD VOLUME

⬇

↑ATRIAL STRETCH RECEPTORS STIMULATED

⬇

B-FIBERS IN ATRIA ACTIVATED AND SENDS SIGNALS TO **VASOMOTOR CENTER** (**VMC**)

⬇

↑ **SYMPATHETIC ACTIVITY**, <u>INHIBIT</u> PARASYMPATHETICS

⬇

↑ HR

Atrial Stretch and ANP

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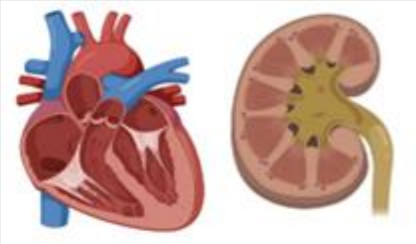
Atrial stretch causes a **non-neural** response from the stretching of the atrial cardiomyocytes → Secretion of **Atrial Natriuretic Peptide (ANP)**

- **ANP** acts via **cGMP**

ANP Effects:

- Causes **vasodilation**
- **↓Na+ reabsorption** at renal collecting tubule
- **Dilates afferent renal arterioles** & **constricts efferent arterioles** → **Diuresis**



MECHANISM

↑BLOOD VOLUME

↓

↑ATRIAL CARDIOMYOCYTES STRETCH

↓

↑ANP SECRETION → **↑ RELAXATION** OF **VASCULAR SMOOTH MUSCLE CELLS** → **VASODILATION** & **↓TPR** → **↓BP**

↓

- **↑RENAL EXCRETION OF NACL** & **WATER**
- **↓NA+ REABSORPTION** AT RENAL COLLECTING TUBULE
- INHIBITION OF RENIN
- **DILATES AFFERENT** ARTERIOLES
- CONSTRICTS EFFERENT ARTERIOLES

→ **DIURESIS** → **↓BP**

REVIEW QUESTIONS

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- **Where are the baroreceptors located?**
- **Which cranial nerves are involved in baroreceptor reflex?**
- **Which part of the medulla receives afferent signals involved in baroreceptor reflex?**
- **Briefly describe baroreceptor reflex in the case of hypotension.**
- **Explain how ↑BP and ↓HR occur in Cushing reflex.**

REVIEW QUESTIONS - ANSWERS

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Where are the baroreceptors located?

- Carotid sinus
- Aortic arch

Which cranial nerves are involved in baroreceptor reflex?

- **CN IX** (*Glossopharyngeal n*. in carotid sinus)
- **CN X** (*Vagus n.* in aortic arch)

Which part of medulla receives afferent signals involved in baroreceptor reflex?

- Nucleus tractus solitarius (NTS)

Briefly describe baroreceptor reflex in the case of hypotension.

[↓ BP]→ [↓ Stretch detected by baroreceptors] → [↓ Baroreceptor firing rate] → [NTS receives signals and promotes sympathetic stimulation & decreases parasympathetic activity]↓ → [↑ HR, ↑ SV, ↑ BP]

Explain how ↑BP and ↓HR occur in Cushing reflex.

[↑ ICP]→ [Cerebral ischemia] → [↑ pCO_2 and ↓ pH] → [Compensatory activation of sympathetics] → [↑ BP] → [Baroreceptors detect ↑ BP] → [↑ Baroreceptor firing rate] → [↓ HR]

PRACTICE QUESTION 1 - ANSWER

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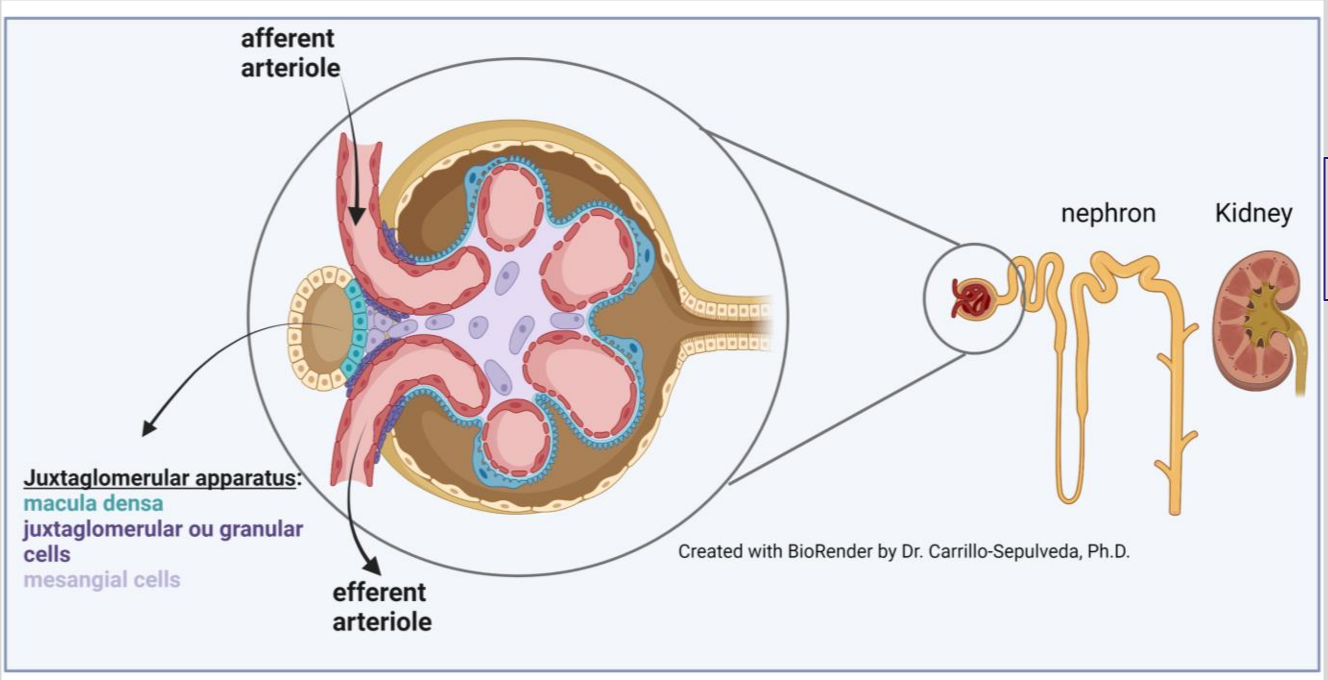
If the vagus nerve (CN X) were damaged, baroreceptor input in the aortic arch would not be received by the NTS.

However, in contrast to baroreceptors in the carotid sinuses, the aortic arch baroreceptors are not very good at sensing low BP, thus, it would not make the brain think that there is hypotension.

Severing the vagus nerve would remove parasympathetic stimulation to the heart. Therefore, with the unopposed sympathetic stimulation in the heart, the HR would increase.

RENAL BP REGULATION - JUXTAGLOMERULAR APPARATUS



MECHANISM

↓**Renal perfusion pressure (detected by baroreceptors in afferent arteriole)**

↓**NaCl (hyponatremia)** detected by **macula densa**

JG cells secrete renin

↓

↑**Renin release**

↓

↑Conversion of angiotensinogen to **angiotensin I**

↓

↑Conversion of **angiotensin I** to **angiotensin II**

[Release of renin from the juxtaglomerular cells]→ [RAAS activation] → **[Direct <u>vasoconstriction</u>** and ↑ extracellular volume through↑ **<u>sodium</u> and water reabsorption, ↓ K+, ↑ <u>pH</u>)]**

ACTIVATION OF SYSTEMIC RAAS

↓MAP

↓Renal perfusion pressure

Stretch receptors in JG cells sense ↓renal perfusion pressure

↑**Renin** release by granular* cells into the bloodstream

RAAS activated

↑**Arteriolar vasoconstriction**

↑**TPR**

↑**Extracellular fluid** & **blood volume**

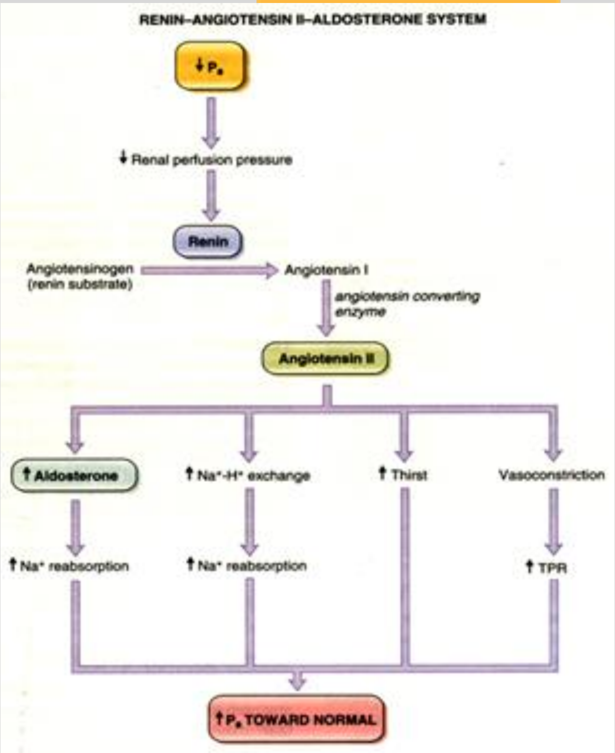
↑**Venous return**

↑**CO**

↑ **MAP**

MAP = CO x TPR

*__Juxtaglomerular (JG) cells__ are also known as **granular cells**.



Costanzo, Physiology 5th ed., Saunders/Elsevier, 2014 (Chapter 4: pp. 160-161)

CLINICAL RELEVANCE

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Hypertension (HTN)

Risk Factors:

- ↑ Age
    - ♂>♀ Below age 65
    - ♀>♂ after menopause
- Obesity
- Diabetes
- High-sodium diet
- Excess EtOH
- Tobacco smoking
- Sedentarism
- Family history
- Higher rates in African American > White > Asian, Hispanic

BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)	and/or	DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120 – 129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130 – 139	or	80 – 89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

American Heart Association, 2017

Primary HTN:
~90% of cases in adults, increasing prevalence in children and adolescents

- **Related to ↑CO or ↑TPR**

Secondary HTN:
~10% of cases

- May be secondary to **renal or renovascular diseases**, such as fibromuscular dysplasia and atherosclerotic renal artery stenosis.

HTN → ↑↑ risk for many diseases, such as:

- **CAD**
- **Heart Failure**
- **Atrial fibrillation**
- **CVA/TIA**
- **CKD**

HEMORRHAGE

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Low blood volume → ↓ BP

BARORECEPTOR REFLEX

Baroreceptors detect ↓ stretch

↓ Firing frequency of baroreceptors

Afferent signals sent via **CN IX** & **CN X**

Signals received by **vasomotor center** in medulla (**NTS)**

↓ **Parasympathetic** stimulation

↑ HR

↑ **Sympathetic** stimulation

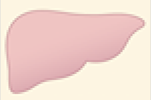
↑ Force of contraction

Vasoconstriction

↑ Peripheral Resistance

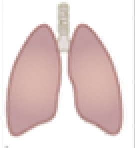
KIDNEY & RAAS

+ Renin



+ Angiotensinogen

Angiotensin I



+ ACE

Angiotensin II

↑Sympathetic activity

↑Arteriolar vasoconstriction

↑Renal Na+/Cl- resorption

↑Aldosterone secretion

↑ADH secretion

↑Salt & water retention

	CAUSED BY	SKIN	PCWP (PRELOAD)	CO	SVR (AFTERLOAD)	TREATMENT
Hypovolemic shock	Hemorrhage, dehydration, burns	Cold, clammy	↓↓	↓	↑	IV fluids

Le T, Sochat M, Schimansky S, et al. *First Aid For the USMLE Step 1*. New York: McGraw Hill; 2021.

REVIEW QUESTIONS

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- **Where are the renal baroreceptors located?**
- **How does the kidney contribute to BP regulation and what is the net effect?**
- **Which type of renal cells secrete renin?**
- **Name 3 factors that trigger renin secretion.**
- **Briefly describe the role of renin in RAAS and BP regulation in response to hypotension.**

REVIEW QUESTIONS - ANSWERS

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Where are the renal baroreceptors located?

- **Afferent arterioles**

How does the kidney contribute to BP regulation and what is the net effect?

- **Through secretion of renin, resulting in changes in salt and water retention**

Which type of renal cells secrete renin?

- **Juxtaglomerular (JG) cells**, also known as **granular cells**

Name 3 factors that trigger renin secretion.

- ↓ **Renal perfusion pressure** (detected by **baroreceptors** in **afferent arteriole**)
- ↑**Renal sympathetic discharge**
- ↓ **NaCl delivery to macula densa cells.**

Briefly describe the role of renin in RAAS and BP regulation in response to hypotension.

[↓ BP]→ [↓ Renal perfusion pressure, ↓NaCl delivery to macula densa cells] → [↑ Secretion of renin by JG cells] → [↑Conversion of angiotensinogen to angiotensin I by renin] → [↑Conversion of angiotensin I to angiotensin II by ACE] → [↑Angiotensin II activity] → [↑Sympathetic activity, arteriolar vasoconstriction, renal NaCl reabsorption, aldosterone secretion, ADH secretion] → **[↑Salt & water retention] → [↑ BP]**